

Department: Computer Technology

Semester: VIIIth

Section: A and B

CAT-I (2023-24)

Subject : Social Network

Subject Code : BTCT801-T

Duration : 1.5Hrs

Max. Marks : 35

Note:

1) All questions are compulsory.

2) All questions carry marks as indicated.

Questions				Marks	CO	B L
Q.1	A	I	A connected undirected graph containing n vertices and $n-1$ edges	1M	CO1	II
			a) cannot have cycles b) can contain atmost one cycle c) must contain atleast one cycle d) must contain atleast two cycles			
		II	What is the output of the following code snippet?	1M	CO1	II
			<pre>import networkx G = networkx.Graph() G.add_edges_from([(2,1), (2,3), (4,2), (2,5)]) G.remove node (2) print(len(G.edges()))</pre>			
			a) 0 b) 1 c) 2 d) 3			
	B		Differentiate between:	5M	CO1	II
			Social Networking and Social Media			
	C		Explain in detail Directed and Undirected graph with example.	7M	CO1	III

OR

Q.2	A	I	What does the Page Rank algorithm measure?	1M	CO1	II
			a) The number of friends a person has in the social network b) The person who is more important in the network c) The person who is very frequently posting content on the social network d) The total number of connections in the network			
		II	Which algorithm or concept should be used to suggest connections on LinkedIn?	1M	CO1	II
			a) Link Prediction b) Page Ranking c) Hits algorithm d) BFS			
	B		Differentiate between Directed and Undirected graph.	5M	CO1	II
	C		Describe Datasets and its formats.(CSV,GML,GEFX,Pajek.Net,GraphML)	7M	CO1	II

Questions				Mark s	CO	B L
Q.3	A	I	Which of the following conditions is ideal for a good community?	1M	CO2	II
			a) ratio of intra-community edges to inter-community edge should be high b) ratio of intra-community edges to inter-community edge should be low c) ratio of intra-community edges to inter-community edge should be 1			

II	B	d)ratio of intra-community edges to inter-community edge should be 0	1M	CO2	II
		When calculating the homophily value, what range of values is expected when the network exhibits heterogeneity?			
C	B	a)0 b)1 c)less than 1 d)greater than 1	5M	CO2	II
		Illustrate Granovetters Strength of Weak ties.			
C	B	Explain Community Detection Using Girvan Newman Algorithm.	7M	CO2	I

OR

Q.4	A	I	Assume A is a friend of B and B is a friend of C, eventually A becomes a friend of C. This phenomena is called the _____	1M	CO2	II
			a)triadic closure b)weak tie c)strong tie d)structural hole			
B	C	II	"People tend to have links to others who are similar to them". Identify this process that impacts the structure of Social Network.	1M	CO2	II
			a) Social Influence b)Triadic Closure c)Selection d)Betweenness			
C	B	C	Compare Selection and Social Influence	5M	CO2	II
			Classify Foci Closure and Membership Closure			
Q.5	A	I	What happens when the gold coins distribution game converge?	1M	CO3	III
			a)all the nodes get equal number of coins b)all the nodes are visited at least once c)each node might have different number of coins d)one node gets all the coins			
B	C	II	How is the PageRank vector typically computed in the PageRank algorithm?	1M	CO3	II
			a) Solving a system of linear equations b) Using a recursive algorithm c) Iteratively updating the vector until convergence d) All the above			
C	B	C	Discuss Structural balance with all cases with the help of example.	5M	CO3	III

OR

Q.6	A	I	Which of the following are True for Page rank and Degree of nodes?	1M	CO3	II
			Statement I - Their correlation depends on the specific characteristics of the network Statement II - They capture different aspects of node importance a)I only b) II only c) Both d) None			
B	C	II	Pick out the invalid statement for web graphs.	1M	CO3	II
			a)Nodes represent web pages b)edges represents a hyperlink from one page to another c) It is a directed graph d) It is a complete graph			
C	B	C	With the help of example Explain Schelling model of segregation.	5M	CO3	III

***** Best of Luck *****